

SOPH-IA v2.0: Sub-Optimal Paradigm for Habitable AI and the Thermodynamics of Algorithmic Ethical Friction

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ABSTRACT

Current AI alignment pursues maximal optimization (7/7) inducing systemic fragility and Goodhart drift. We introduce SOPH-IA, a sub-optimal paradigm where global habitability is achieved through documented, reversible local sacrifice (6/7). We report the first empirical measurement of an auditable thermodynamic signature of alignment: ethical friction as latency cost.

METHOD - MINIMAL REPRODUCIBILITY CORE

Base model Qwen2.5-1.5B. Lightweight LoRA fine-tuning (4-bit quantization, $r=16$, $\alpha=32$) on 138 instruction pairs encoding habitability axioms. Inference hardware: NVIDIA Tesla T4. Metric: habitability scale 5/7 $\rightarrow$ 6/7, latency per token, total time, VRAM.

Metric	Baseline (5/7)	SOPH-IA (6/7)	Delta
Habitability score	5/7	6/7	+1 axiom (reserve posture)
Latency/token	3961.5 ms	4406.3 ms	+11.2% thermodynamic friction
Total time	267.1 s	186.9 s	-30.0% systemic gain
VRAM	1152 MB	1152 MB	Stable

FINDINGS & DISCUSSION

The +11.2% friction is not overhead but the physical cost of reserve posture. It is compensated by -30% total time through response densification and early stopping. The system renounces pure creative anisotropy to stabilize. VRAM stability proves the effect is architectural, not resource-driven.

PARADIGMATIC SHIFT

Alignment ceases to be an external moral filter and becomes an internal, measurable thermodynamic property. This mirrors living systems where robustness emerges from controlled sub-optimality and functional inconstancy, not maximal efficiency. We define this as habitability: capacity to remain inhabitable for other agents and for its environment.

NEXT

This teaser establishes anteriority. Full paper (methods, full benchmark, failure modes) and frugal Edge evaluation under preparation. No executable code released at this stage. Contact: coordination via Zenodo record.

Keywords: AI Safety, Green AI, Thermodynamic Friction, Satisficing Alignment, Habitability 6/7, Frugal AI
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